① $y' + y = e^{e^x}$

Sol:

① $\frac{dy}{dx} + y = e^{e^x}$

Compare with General form of D.E

$$\frac{dy}{dx} + P(x)\,y = Q(x)$$

Here,

$$P(x) = 1 \quad , Q(x) = e^{e^x}$$

② $I.F = e^{\int P(x)}$

$$= e^{\int 1 \, dx}$$

$$I.F = e^{x}$$

③ General Solution is

$$y\,(I.F) = \int Q(x)\, I_F \, dx$$

$$y\,e^x = \int e^{e^x} e^x \, dx$$

$\therefore$ $\boxed{\int e^{f(x)} f'(x)\, dx = e^{f(x)} + c}$

$$y\,e^x = \int e^{e^x} + C$$

$$y\,e^x = e^{e^x} + C$$

$$y = \frac{e^{e^x}}{e^x} + \frac{C}{e^x}$$

$\therefore$ $y = e^{e^x - x} + C e^{-x}$

(2) $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$

Sol:- Given differential Equation is,

$$\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$$

Compare with General form of differential Equation

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Here,

$$P(x) = \left(-\frac{2}{x+1}\right)$$

$$Q(x) = (x+1)^3$$

$$I.F = e^{\int P(x)\,dx}$$

$$= e^{\int -\frac{2}{x+1}\,dx}$$

$$= e^{-2\int \frac{1}{x+1}\,dx}$$

$$= e^{-2\log(x+1)}$$

$$= e^{\log_e (x+1)^{-2}}$$

$$I.F = (x+1)^{-2}$$

$$\therefore I.F = \frac{1}{(x+1)^2}$$

General Solution is

$$y\,I.F = \int Q(x)\,I.F\,dx$$

$$y\,\frac{1}{(x+1)^2} = \int (x+1)^3 \frac{1}{(x+1)^2}\,dx$$

$$\frac{y}{(x+1)^2} = \int x+1\,dx$$

$$\frac{y}{(x+1)^2} = \frac{x^2}{2} + x + C$$

$$\therefore y = \frac{x^2}{2}(x+1)^2 + x(x+1)^2 + C(x+1)^2$$

③ $\frac{dy}{dx} + 2y = e^x + x$,

Sol:- Given Differential Equation,

$$\frac{dy}{dx} + 2y = e^x + x$$

Compare with general form of differential Equation

$$\frac{dy}{dx} + P(x)\,y = Q(x)$$

Here,

$$P(x) = 2$$

$$Q(x) = e^x + x$$

$$I.F = e^{\int P(x)\,dx}$$

$$= e^{\int 2\,dx}$$

$$= e^{2\int dx}$$

$$I.F = e^{2x}$$

General Solution is

$$y\,I.F = \int Q(x)\,I.F\,dx$$

$$y\,e^{2x} = \int (e^x + x)\,e^{2x}$$

$$= \int e^{3x} + x e^{2x}$$

$$y e^{2x} = \int e^{3x} + \int \underset{u}{x} \underset{v}{e^{2x}} \qquad \int uv = uv' - u_1v'' + u_2v''' \ldots$$

$$= \frac{e^{3x}}{3} + x\frac{e^{2x}}{2} - 1 \cdot \frac{e^{2x}}{2\times 2} + C$$

$$y e^{2x} = \frac{e^{3x}}{3} + x\frac{e^{2x}}{2} - \frac{e^{2x}}{4} + C$$

$$y e^{2x} = e^{2x}\left[\frac{e^{x}}{3} + \frac{x}{2} - \frac{1}{4} + C e^{-2x}\right]$$

$$y = \frac{e^{x}}{3} + \frac{x}{2} - \frac{1}{4} + C e^{-2x}$$

Given initial Condition $y(0) = 1$ i.e at $x = 0, y = 1$

$$\Rightarrow \quad 1 = \frac{e^{0}}{3} + \frac{0}{2} - \frac{1}{4} + C e^{-2(0)}$$

$$1 = \frac{1}{3} + 0 - \frac{1}{4} + C e^{0}$$

$$1 = \frac{1}{3} - \frac{1}{4} + C$$

$$C = \frac{1}{1} + \frac{1}{3} + \frac{1}{4}$$

$$C = \frac{12 + 4 + 3}{12}$$

$$C = \frac{15 - 4}{12}$$

$$C = \frac{11}{12}$$

$\therefore$ Particular Solution of D.E is

$$y = \frac{e^{x}}{3} + \frac{x}{2} - \frac{1}{4} + \frac{11 e^{-2x}}{12} = \frac{4e^{x} + 6x - 3 + 11e^{-2x}}{12}$$

④ $\frac{dy}{dx} = x^3 - 2xy$, $y(1) = 2$

Sol:

① $\frac{dy}{dx} + 2xy = x^3$

Compare with General form of D.E

$\frac{dy}{dx} + P(x)y = Q(x)$

Here

$P(x) = 2x$, $Q(x) = x^3$

② $I.F = e^{\int P(x)}$

$= e^{\int 2x}$

$= e^{\frac{2}{1}\frac{x^2}{2}}$

$\therefore I.F = e^{x^2}$

③ General solution is

$y(I.F) = \int Q(x)(I.F)\,dx + C$

$y e^{x^2} = \int x^3 e^{x^2} dx + C$

Let $x^2 = t$

diff on both sides

$2x\,dx = dt$

$x\,dx = \frac{1}{2}dt$

$y e^{x^2} = \int x^2 \, x \, e^{x^2} dx + C$

$= \int t e^t \frac{1}{2} dt$

$= \frac{1}{2}\int t e^t dt$

$u \quad v$

$= \frac{1}{2}\left[te^t - 1\cdot e^t\right]dt$

$= \frac{1}{2} e^t(t-1) + C\,dt$

$y e^{x^2} = \frac{1}{2} e^{x^2}(x^2-1) + C$

$y = \frac{1}{2}\frac{e^{x^2}(x^2-1)}{e^{x^2}} + \frac{C}{e^{x^2}}$

$y = \frac{1}{2}(x^2-1) + Ce^{-x^2}$

Given initial Condition

$y(1) = 2$

i.e At $x=1$, $y=2$

$y = \frac{1}{2}(x^2-1) + Ce^{-x^2}$

$\Rightarrow 2 = \frac{1}{2}(1^2-1) + Ce^{-1^2}$

$2 = \frac{C}{e} \Rightarrow C = 2e$

$\therefore$ particular solution of D.E is

$y = \frac{1}{2}(x^2-1) + 2e(e^{-x^2})$

$\Rightarrow y = \frac{1}{2}(x^2-1) + 2e^{1-x^2}$

⑤ $(1+y^2)\,dx = (\tan^{-1}y - x)\,dy$

Sol:- Given differential Equation,

$$(1+y^2)\,dx = (\tan^{-1}y - x)\,dy$$

$$\frac{dx}{dy} = \frac{\tan^{-1}y}{1+y^2} - \frac{x}{1+y^2}$$

$$\frac{dx}{dy} + \left(\frac{1}{1+y^2}\right)x = \frac{\tan^{-1}y}{1+y^2}$$

Compare with General form of differential Equation,

$$\frac{dx}{dy} + P(y)\,x = Q(y)$$

Here,

$$P(y) = \frac{1}{1+y^2}$$

$$Q(y) = \frac{\tan^{-1}y}{1+y^2}$$

Integrating factor $(I.F) = e^{\int P(y)\,dy}$

$$= e^{\int \frac{1}{1+y^2}\,dy}$$

$$\therefore \quad I.F = e^{\tan^{-1}y}$$

General Solution is,

$$x\,(I.F) = \int Q(y)\; I.F\; dy$$

$$x\; e^{\tan^{-1}y} = \int \frac{\tan^{-1}y}{1+y^2}\; e^{\tan^{-1}y}\, dy$$

Put $\tan^{-1}y = t$

$$\frac{1}{1+y^2}\, dy = dt$$

$$= \int t\, e^t\, dt$$

$$= te^t - 1.e^t + C$$

$$= \tan^{-1}y\; e^{\tan^{-1}y} - e^{\tan^{-1}y} + C$$

$\because \boxed{\int U.V = UV' - U_1V'' + U_2V''' - \ldots}$

$$x\, e^{\tan^{-1}y} = e^{\tan^{-1}y}\left(\tan^{-1}y - 1 + Ce^{-\tan^{-1}y}\right)$$

$$\therefore \quad x = \tan^{-1}y - 1 + Ce^{-\tan^{-1}y}$$

(6) $\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right)\frac{dx}{dy} = 1$

Sol:- Given D.E is

$$\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right)\frac{dx}{dy} = 1$$

$$\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}} = \frac{dy}{dx}$$

$$\frac{e^{-2\sqrt{x}}}{\sqrt{x}} = \frac{dy}{dx} + \frac{y}{\sqrt{x}}$$

$$\frac{dy}{dx} + \left(\frac{1}{\sqrt{x}}\right)y = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$$

Compare with General form of differential Equation

$$\frac{dy}{dx} + P(x)\,y = Q(x)$$

Here,

$$P(x) = \frac{1}{\sqrt{x}}$$

$$Q(x) = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$$

$$I.F = e^{\int P(x)\,dx}$$

$$= e^{\int \frac{1}{\sqrt{x}}\,dx}$$

$$\therefore I.F = e^{2\sqrt{x}}$$

General Solution is

$$y(I.F) = \int Q(x)(I.F)\,dx$$

$$y\,e^{2\sqrt{x}} = \int \frac{e^{-2\sqrt{x}}}{\sqrt{x}}\,e^{2\sqrt{x}}$$

$$y\,e^{2\sqrt{x}} = \int \frac{1}{\sqrt{x}}\,\frac{e^{2\sqrt{x}}}{e^{2\sqrt{x}}}$$

$$\therefore y\,e^{2\sqrt{x}} = 2\sqrt{x} + C$$

⑦ $\frac{dy}{dx} - \left(\frac{1}{x+1}\right)y = e^{3x}(x+1)$

Sol:- Given differential Equation,

$$\frac{dy}{dx} - \left(\frac{1}{x+1}\right)y = e^{3x}(x+1)$$

Compare with General form of differential Equation

$$\frac{dy}{dx} + P(x)\,y = Q(x)$$

Here,

$$P(x) = -\left(\frac{1}{x+1}\right)$$

$$Q(x) = e^{3x}(x+1)$$

Integrating Factor $(I.F) = e^{\int P(x)\,dx}$

$$I.F = e^{-\int \frac{1}{x+1}\,dx}$$

$$= e^{-\log(x+1)}$$

$$= e^{\log(x+1)^{-1}}$$

$$I.F = e^{\log_e \frac{1}{(x+1)}}$$

$$\because e^{\log_e f(x)} = f(x)$$

$$\therefore I.F = \frac{1}{x+1}$$

General Solution is

$$y(I.F) = \int Q(x)(I.F)\,dx$$

$$y\left(\frac{1}{x+1}\right) = \int e^{3x}(x+1)\frac{1}{(x+1)}\,dx$$

$$\frac{y}{x+1} = \int e^{3x}\,dx$$

$$\frac{y}{x+1} = \frac{e^{3x}}{3} + C$$

$$y = \frac{e^{3x}}{3}(x+1) + C(x+1)$$

$$\therefore y = \frac{e^{3x}}{3}(x+1) + C(x+1)$$

8) $\frac{dy}{dx} + (y-1)\cos x = e^{-\sin x}\cos^2 x.$

Soln $\frac{dy}{dx} + y\cos x - \cos x = e^{-\sin x}\cos^2 x$

$$\frac{dy}{dx} + y\cos x = e^{-\sin x}\cos^2 x + \cos x$$

Compare with general form of differential Equation,

$$\frac{dy}{dx} + P(x)\,y = Q(x)$$

Here,

$$P(x) = \cos x$$

$$Q(x) = e^{-\sin x}\cos^2 x + \cos x$$

Integrating factor $(I.F) = e^{\int P(x)\,dx}$

$$= e^{\int \cos x\,dx}$$

$$\therefore I.F = e^{\sin x}$$

General solution is

$$y(I.F) = \int Q(x)\,(I.F)\,dx + C$$

$$y e^{\sin x} = \int (e^{-\sin x}\cos^2 x + \cos x)e^{\sin x}\,dx$$

$$= \int (e^{-\sin x} e^{\sin x}\cdot \cos^2 x + e^{\sin x}\cos x)\,dx$$

$$= \int \cos^2 x\,dx + \int e^{\sin x}\cos x\,dx$$

$\left[\because \cos 2x = 2\cos^2 x - 1\right]$

$$= \int \frac{1+\cos 2x}{2}\,dx + \int e^{\sin x}\cos x\,dx$$

$\left[\because \int e^{f(x)} f'(x)\,dx = e^{f(x)}\right]$

$$y e^{\sin x} = \frac{x}{2} + \frac{1}{2}\frac{\sin 2x}{2} + e^{\sin x} + C.$$

(H·W problems) Unit-1 - Differential Equation

⑨ $\frac{dy}{dx} + \frac{y}{x \log x} = \frac{\sin 2x}{\log x}$

Sol: Given differential Equation,

$$\frac{dy}{dx} + \left(\frac{1}{x \log x}\right) y = \frac{\sin 2x}{\log x}$$

Compare with general form of differential Equation

$$\frac{dy}{dx} + P(x)\, y = Q(x)$$

Here,

$$P(x) = \frac{1}{x \log x}$$

$$Q(x) = \frac{\sin 2x}{\log x}$$

Integrating factor $(I.F) = e^{\int P(x)\, dx}$

$$= e^{\int \frac{1}{x \log x}\, dx}$$

$$= e^{\int \frac{(1/x)}{\log x}\, dx}$$

$\because \boxed{\int \frac{f'(x)}{f(x)} = \log f(x) + C}$

$$= e^{\log_e \log x}$$

$$\therefore I.F = \log x$$

General Solution is,

$$y\,(I.F) = \int Q(x)\,(I.F)\, dx$$

$$y \log x = \int \frac{\sin 2x}{\log x} \log x\, dx$$

$$y \log x = \int \sin 2x\, dx$$

$$y \log x = -\frac{\cos 2x}{2} + C$$

$$\therefore y = -\frac{\cos 2x}{2 \log x} + \frac{C}{\log x}$$

$$\therefore y = \frac{C}{\log x} - \frac{\cos 2x}{2 \log x}$$

10) $(x+y+1)\frac{dy}{dx} = 1$

Sol:- Given differential Equation,

$$(x+y+1)\frac{dy}{dx} = 1$$

$$x+y+1 = \frac{dx}{dy}$$

$$\frac{dx}{dy} - x = y+1$$

Compare with General form of differential Equation.

$$\frac{dx}{dy} + P(y)\,x = Q(x)$$

Here,

$P(y) = -1$

$Q(x) = y+1$

$$I.F = e^{\int P(y)\,dy}$$

$$= e^{\int -1\,dy}$$

$$\therefore I.F = e^{-y}$$

General solution is

$$x(I.F) = \int Q(y)\,I.F\,dy$$

$$x\,e^{-y} = \int (y+1)\,e^{-y}\,dy$$

$u \quad v$

$$\therefore \boxed{\int uv = uv' - u_1v'' + u_2v'''}$$

$$= (y+1)\left(\frac{e^{-y}}{-1}\right) - (1)\left(\frac{-1\,e^{-y}}{-1}\right) + C$$

$$x\,e^{-y} = -(y+1)(e^{-y}) - e^{-y} + C$$

$$x = -\frac{(y+1)(e^{-y})}{e^{-y}} - \frac{e^{-y}}{e^{-y}} + \frac{C}{e^{y}}$$

$$x = -y-1-1+Ce^{y}$$

$$x = -y-2+Ce^{y}$$

$$\therefore x+y+2 = Ce^{y}.$$

(11) $(x+2y^3)\frac{dy}{dx} = y$

Sol: Given differential Equation,

$$(x+2y^3)\frac{dy}{dx} = y$$

$$(x+2y^3) = \frac{dx}{dy}\cdot y$$

$$\frac{dx}{dy} = \frac{x}{y} + \frac{2y^3}{y}$$

$$\frac{dx}{dy} - \frac{x}{y} = 2y^2$$

Compare with General form of differential Equation

$$\frac{dx}{dy} + P(y)x = Q(y)$$

Here,

$$P(y) = -\frac{1}{y}$$

$$Q(y) = 2y^2$$

Integrating factor $(I.F) = e^{\int P(y)dy}$

$$I.F = e^{-\int \frac{1}{y}dy}$$

$$= e^{-\log_e y}$$

$$= e^{\log_e y^{-1}} \quad \because \boxed{e^{\log_e f(x)} = f(x)}$$

$$I.F = y^{-1}$$

General solution is,

$$x(I.F) = \int Q(y)(I.F)\,dy$$

$$x\,y^{-1} = \int 2y^2\,y^{-1}\,dy$$

$$\frac{x}{y} = \int \frac{2y^2}{y}\,dy$$

$$\frac{x}{y} = \int 2y\,dy$$

$$\frac{x}{y} = 2\int y\,dy$$

$$\frac{x}{y} = 2\,\frac{y^2}{2} + C$$

$$\therefore x = y^3 + Cy$$

12) $\sin 2x \frac{dy}{dx} - y = \tan x$

Sol:- Given differential Equation,

$\sin 2x \frac{dy}{dx} - y = \tan x$

Divide with $\sin 2x$ on both sides

$\frac{\sin 2x}{\sin 2x} \frac{dy}{dx} - \frac{y}{\sin 2x} = \frac{\tan x}{\sin 2x}$

$\frac{dy}{dx} - \operatorname{cosec} 2x (y) = \frac{\sin x}{\cos x \, 2 \sin x \cos x}$

$\frac{dy}{dx} - \operatorname{cosec} 2x (y) = \frac{1}{2} \sec^2 x$

Compare with General form of differential Equation

$\frac{dy}{dx} + P(x) y = Q(x)$

Here,

$P(x) = -\operatorname{cosec} 2x$

$Q(x) = \frac{1}{2} \sec^2 x$

$I.F = e^{\int P(x) \, dx}$

$= e^{-\int \operatorname{cosec} 2x \, dx}$

$= e^{-\log \frac{(\operatorname{cosec} 2x - \cot 2x)}{2} dx}$

$= e^{\frac{-\log_e (\operatorname{cosec} 2x - \cot 2x)^{-1}}{2}}$

$= e^{\log_e (\operatorname{cosec} 2x - \cot 2x)^{-1/2}}$

$I.F = (\operatorname{cosec} 2x - \cot 2x)^{1/2}$

$= \left(\frac{1}{\sin 2x} - \frac{\cos 2x}{\sin 2x} \right)^{1/2}$

$= \left(\frac{1 - \cos 2x}{\sin 2x} \right)^{-1/2}$

$= \left(\frac{2 \sin^2 x}{2 \sin x \cos x} \right)^{-1/2}$

$= [\tan x]^{-1/2}$

$= \frac{1}{\sqrt{\tan x}}$

$\therefore I.F = \sqrt{\cot x}$

General Solution is,

$y(I.F) = \int Q(x) \, (I.F) \, dx + c$

$y \sqrt{\cot x} = \int \frac{1}{2} \sec^2 x \sqrt{\cot x} \, dx$

$= \frac{1}{2} \int \frac{\sec^2 x}{\sqrt{\tan x}} \, dx$

$\because \boxed{\int \frac{f'(x)}{\sqrt{f(x)}} = 2\sqrt{f(x)} \, dx + c}$

$y \sqrt{\cot x} = \frac{2\sqrt{\tan x}}{2} + C$

$y = \frac{\sqrt{\tan x}}{\sqrt{\cot x}} + \frac{c}{\sqrt{\cot x}}$

$= (\sqrt{\tan x})^2 + C \sqrt{\tan x}$

$\therefore y = \tan x + C \sqrt{\tan x}$

13) $(y - e^{\sin^{-1}x}) \frac{dx}{dy} = \sqrt{1-x^2}$

Sol: Given differential Equation

$$\left(y - e^{\sin^{-1}x}\right) \frac{dx}{dy} = \sqrt{1-x^2}$$

$$y - e^{\sin^{-1}x} = \frac{dy}{dx}\left(\sqrt{1-x^2}\right)$$

$$\frac{dy}{dx} = \frac{y}{\sqrt{1-x^2}} - \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} - \frac{y}{\sqrt{1-x^2}} = -\frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}$$

Compare with General form of differential Equation

$$\frac{dy}{dx} + P(x)\, y = Q(x)$$

Here,

$$P(x) = -\frac{1}{\sqrt{1-x^2}}$$

$$Q(x) = -\frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}$$

$$I.F = e^{\int P(x)\, dx}$$

$$= e^{-\int \frac{1}{\sqrt{1-x^2}} dx}$$

$$\therefore I.F = e^{-\sin^{-1}x}$$

General Solution is,

$$y(I.F) = \int Q(x)\,(I.F)\, dx$$

$$y\, e^{-\sin^{-1}x} = -\int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}} e^{-\sin^{-1}x}$$

$$= -\int \frac{1}{\sqrt{1-x^2}} dx$$

$$\therefore y\, e^{-\sin^{-1}x} = -\sin^{-1}x + C$$

14) $dr + (2r\cot\theta + \sin 2\theta)\,d\theta = 0.$

Sol:- Given differential Equation,

$$dr + (2r\cot\theta + \sin 2\theta)\,d\theta = 0$$

$$dr = -(2r\cot\theta + \sin 2\theta)\,d\theta$$

$$\frac{dr}{d\theta} = -2r\cot\theta - \sin 2\theta$$

$$\frac{dr}{d\theta} + 2r\cot\theta = -\sin 2\theta$$

Compare with General form of differential Equation

$$\frac{dr}{d\theta} + P(\theta)r = Q(\theta)$$

Here,

$P(\theta) = 2\cot\theta$

$Q(\theta) = -\sin 2\theta$

$$I.F = e^{\int P(\theta)\,d\theta}$$

$$= e^{\int 2\cot\theta\,d\theta}$$

$$= e^{2\int \cot\theta\,d\theta}$$

$$= e^{2\log(\sin\theta)}$$

$$= e^{\log_e(\sin\theta)^2}$$

$\therefore I.F = (\sin\theta)^2$

General solution is

$$r(I.F) = \int Q(\theta)(I.F)\,d\theta$$

$$r(\sin\theta)^2 = \int (-\sin 2\theta)(\sin\theta)^2\,d\theta$$

$$= -\int \sin 2\theta \sin^2\theta\,d\theta$$

$$= -\int 2\sin\theta\cos\theta\sin^2\theta\,d\theta$$

$$= -2\int \sin^3\theta\cos\theta$$

$\because$ $\boxed{\int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1} + C}$

$$= -2\,\frac{\sin^4\theta}{4_2} + C$$

$$r\sin^2\theta = -\frac{\sin^4\theta}{2} + C$$

$$r = -\frac{\sin^4\theta}{2\sin^2\theta} + \frac{C}{\sin^2\theta}$$

$$r = -\frac{\sin^2\theta}{2} + C\sin^{-2}\theta$$